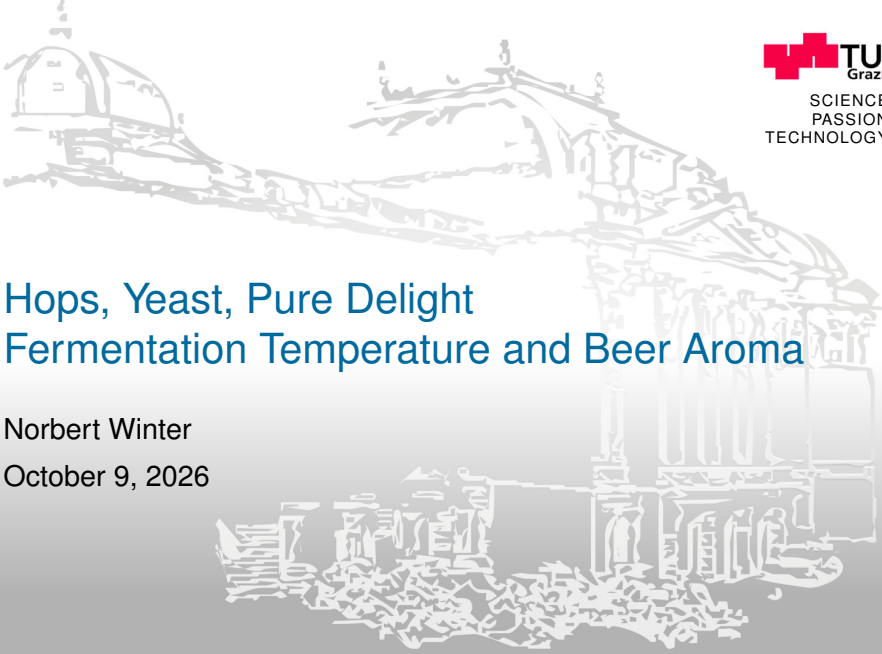




Hops, Yeast, Pure Delight Fermentation Temperature and Beer Aroma

Norbert Winter

October 9, 2026

Outline

1 Motivation

2 Method

3 Results

Why Beer?

- Only four ingredients – yet thousands of aromas
- Much of the aroma is created in the **fermenter**
- Systematic data on fermentation temperature are missing

Research Question

How strongly does fermentation temperature shape the aroma of top-fermented beers?

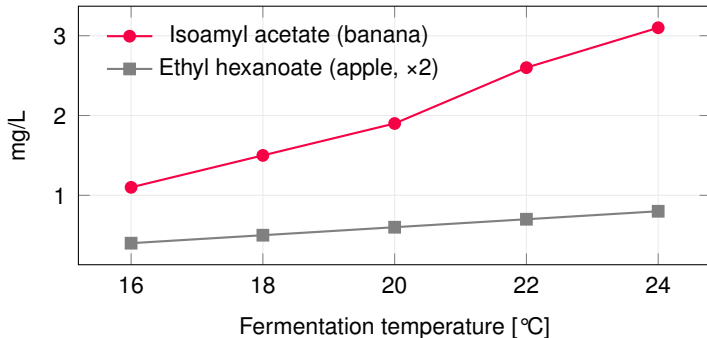
Approach

1. 12 brews of pale ale
2. 4 temperature levels (16–24 °C)
3. Gas chromatography
4. Blind tasting ($n = 24$)

Arrhenius

$$k(T) = A \cdot e^{-E_a/RT}$$

Ester Formation



Tasting

Temperature	Fruitiness	Cleanness	Overall
16 °C	4.1	7.8	6.2
18 °C	5.6	7.2	7.4
20 °C	6.9	5.9	6.8
24 °C	8.2	3.7	5.1

Conclusion

- +4 K $\Rightarrow$ about +38 % isoamyl acetate
- 18 °C is the sweet spot for a pale ale
- Next: kveik yeasts and temperature profiles

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Questions?